

Experiment 8.1

Fourier Transform of Periodic Impulse Train

Program:

```
% Fourier Transform of Periodic Impulse Train

clc;

clear;

close all;

% Time axis
t = -5:0.01:5;

% Generate periodic impulse train
x = zeros(size(t));
x(1:100:end) = 1; % Impulse every 100 samples

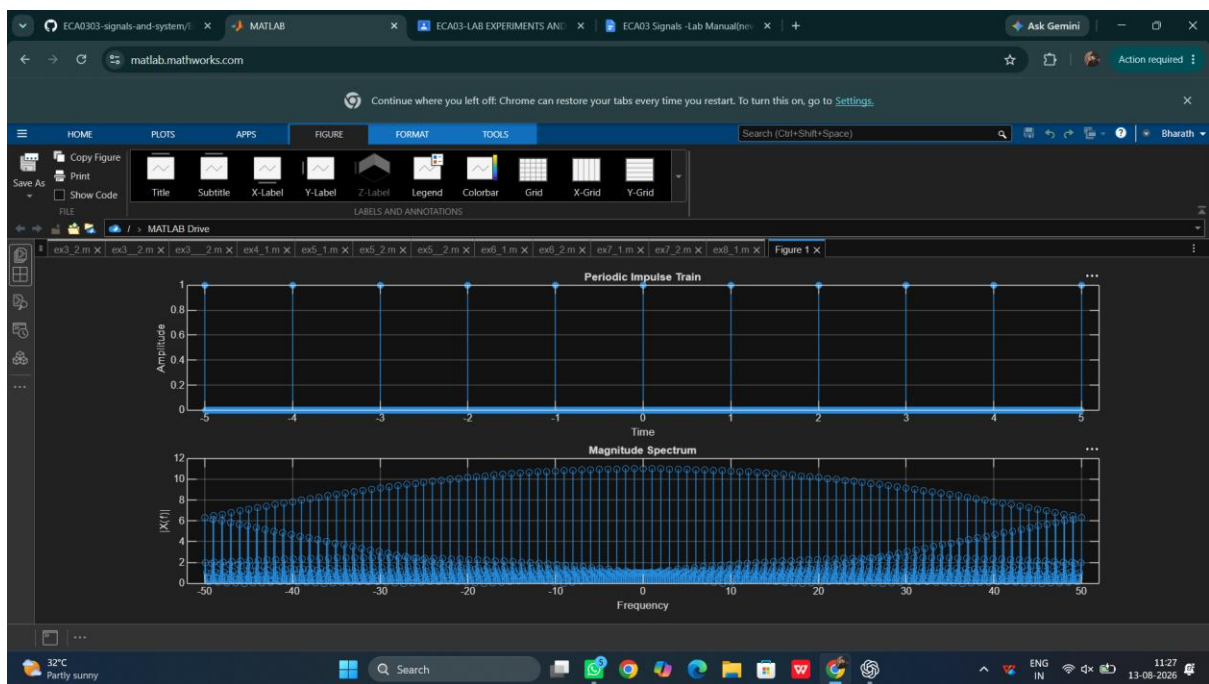
% Fourier Transform
X = fftshift(fft(x));

% Frequency axis
N = length(X);
f = linspace(-50,50,N);

% Plotting
figure;
subplot(2,1,1);
stem(t,x,'filled');
grid on;
title('Periodic Impulse Train');
xlabel('Time');
ylabel('Amplitude');
subplot(2,1,2);
```

```
stem(f,abs(X));  
grid on;  
title('Magnitude Spectrum');  
xlabel('Frequency');  
ylabel('|X(f)|');
```

Output Waveform:



Result:

The Fourier Transform of a periodic impulse train was obtained using MATLAB. The frequency-domain representation was observed to be another impulse train with discrete spectral lines, confirming the Fourier Transform property of periodic impulse trains and illustrating the relationship between time-domain periodicity and frequency-domain discreteness.